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Google Cloud

Let machines do the work

Module 10 - Machine learning
Google Cloud Computing Foundations

Course map

01 — So, what's the cloud anyway?

02 — Start with a solid platform

03 — Use Google Cloud to build your apps

04 — Where do I store this stuff?

05 — There's an API for that!

06 — You can't secure the cloud, right?

07 — It helps to network

08 — Keeping an eye on things

09 — You have the data, but what are you doing with it?

10 — Let machines do the work

Google Cloud skill badges

Objectives

01 Define machine learning (ML), understand the terminology used, and identify the value proposition.

02 Explore Vertex AI, Google's unified AI platform.

03 Use Google APIs to apply a range of pre-trained ML models.

Agenda

01 Machine learning in the cloud

02 Building ML models with Vertex AI

03 Lab: Vertex AI: Qwik Start

04 AutoML

05 Custom training

06 Pre-built APIs

07 Lab: Cloud Natural Language API: Qwik Start

08 Lab: Google Cloud Speech API: Qwik Start

09 Lab: Video Intelligence: Qwik Start

10 Quiz

11 Summary

Agenda

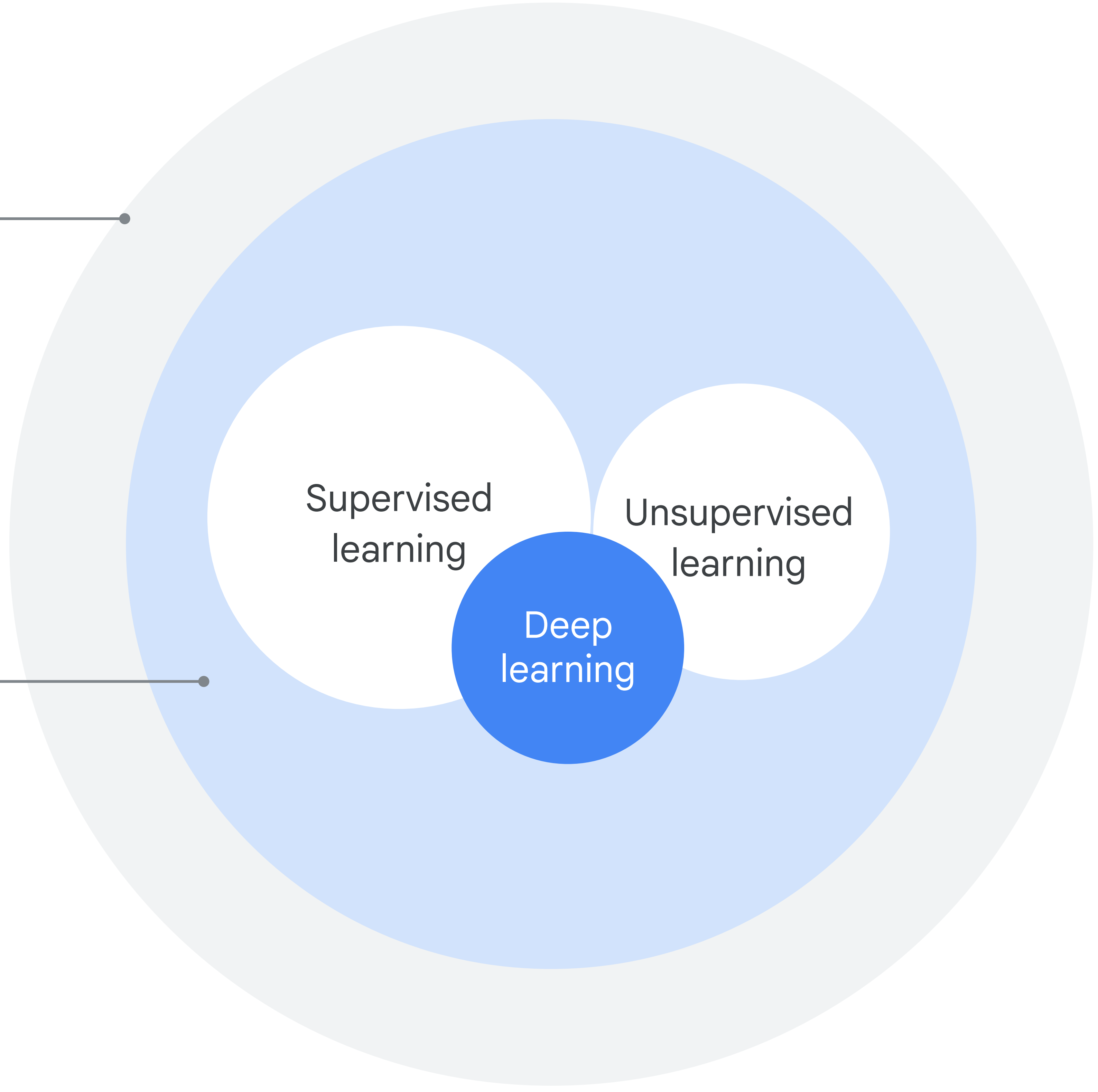
AI vs. ML

Artificial intelligence

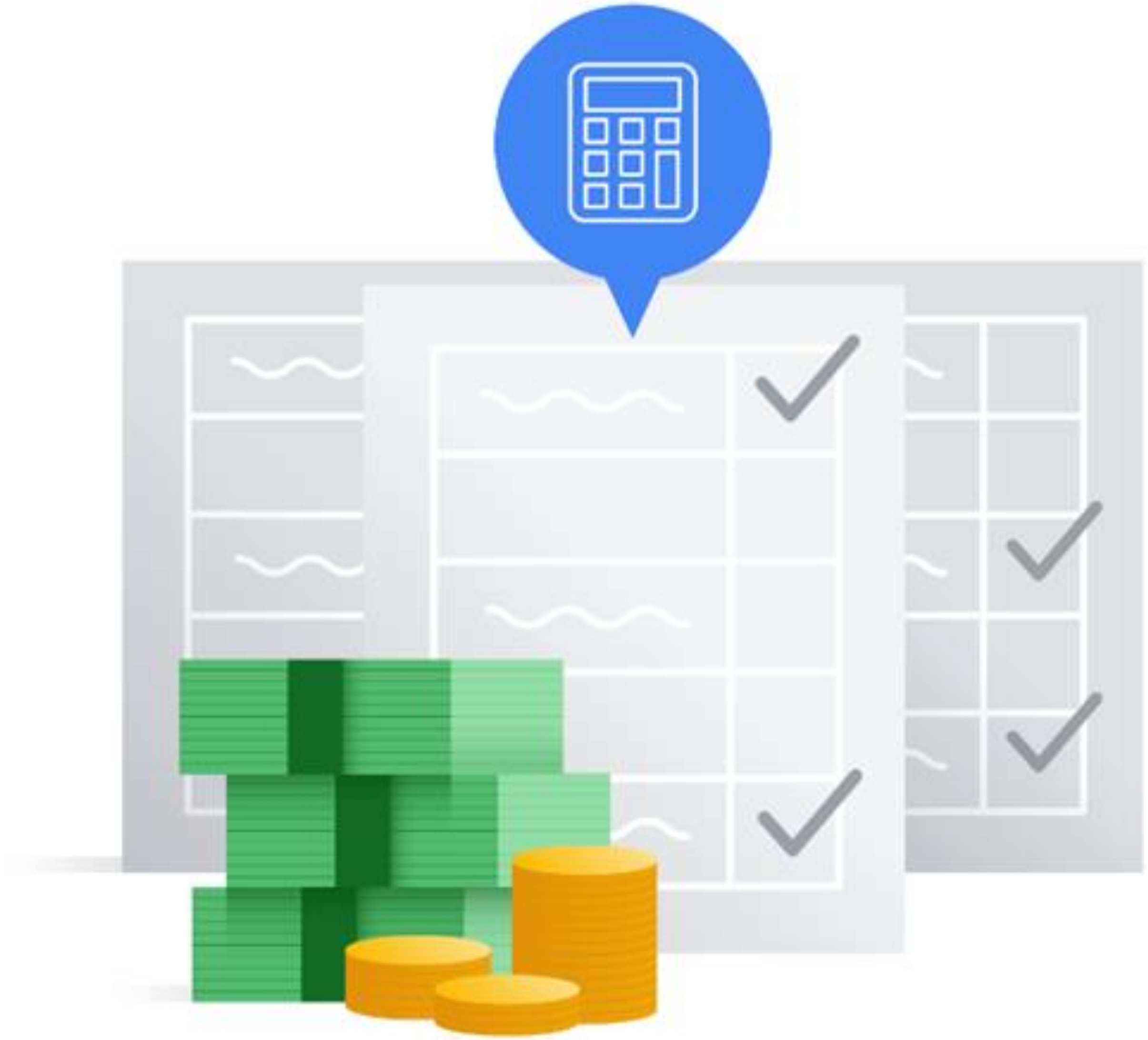
An umbrella term that includes anything related to computers mimicking human intelligence

Machine learning

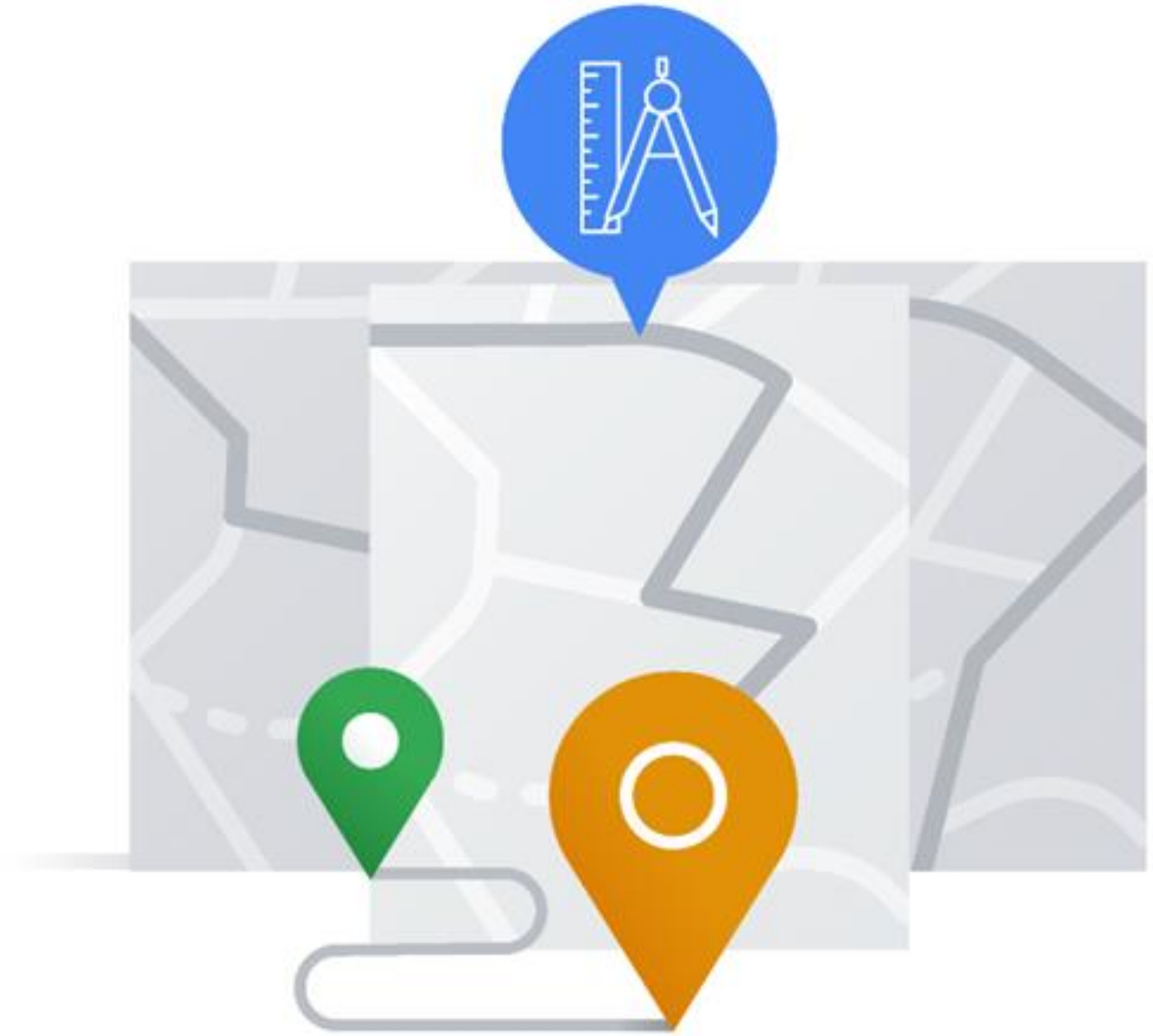
A subset of AI that mainly refers to supervised and unsupervised learning



Model training requires examples

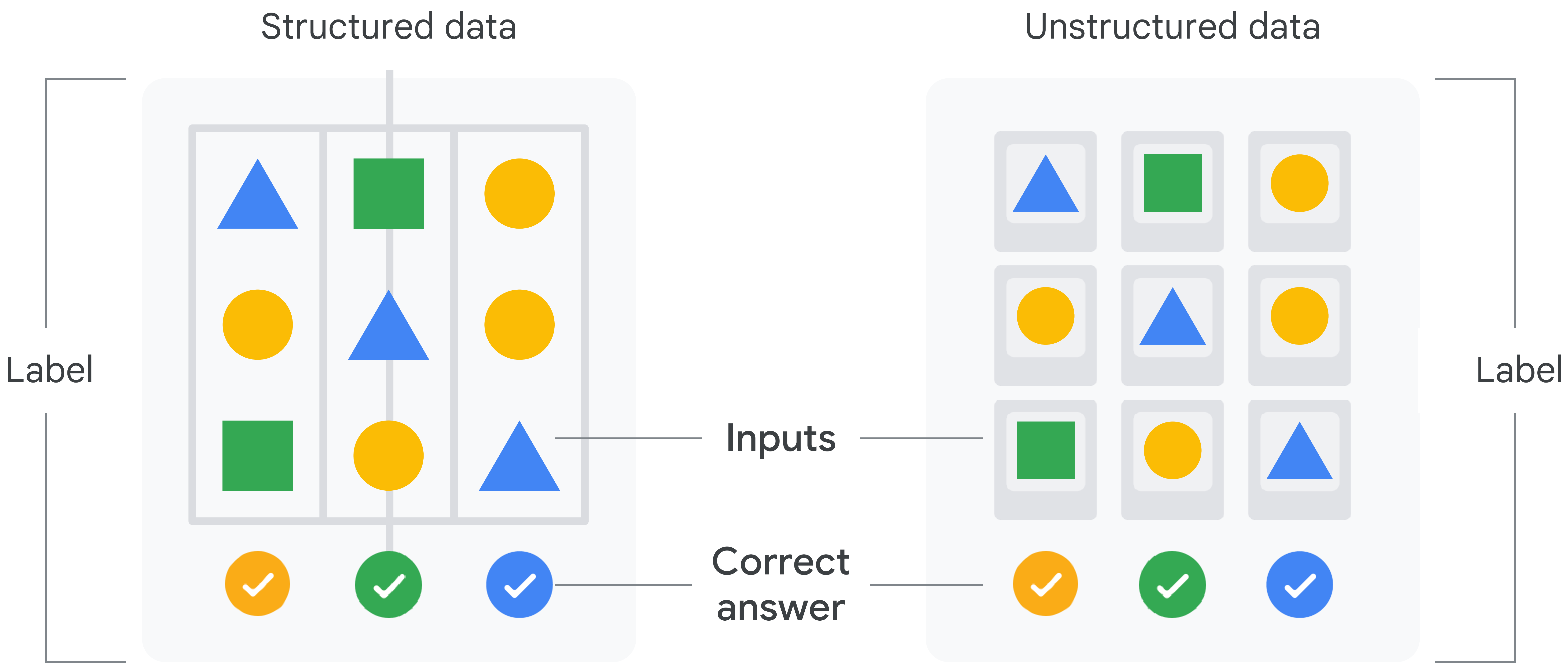


Examples of tax filings

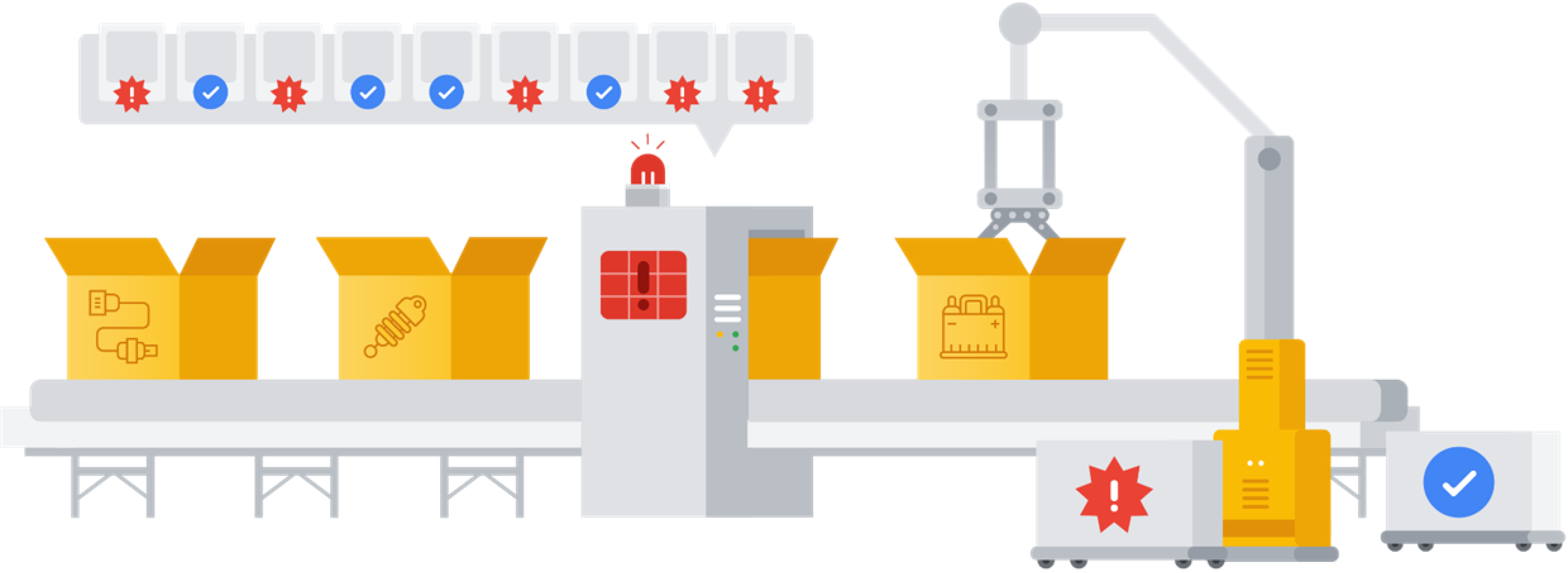


Examples of trips

Train an ML model with examples



Train an ML model with examples

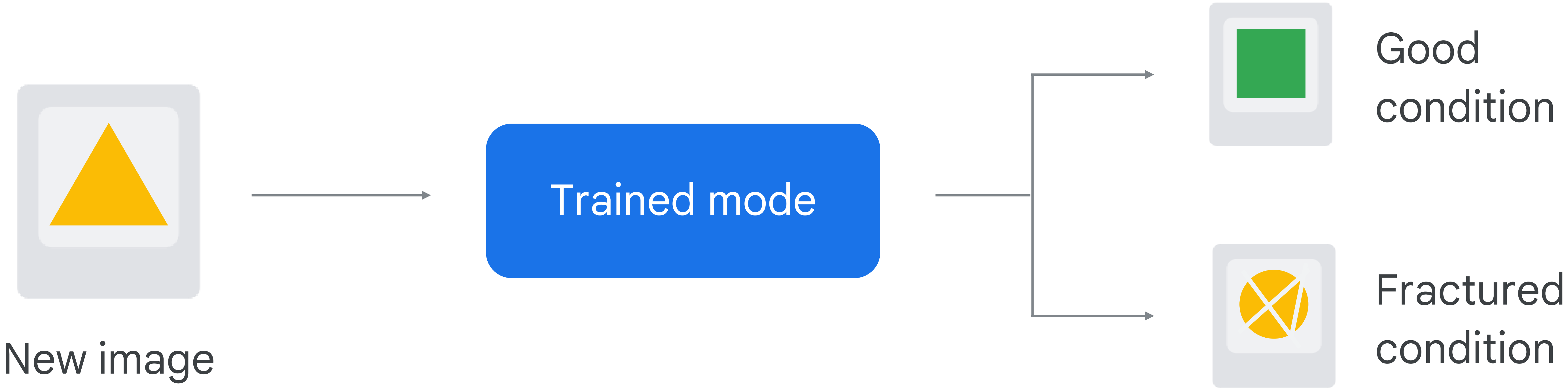


An ML model is only as good as the data used to train it

Parts in good condition

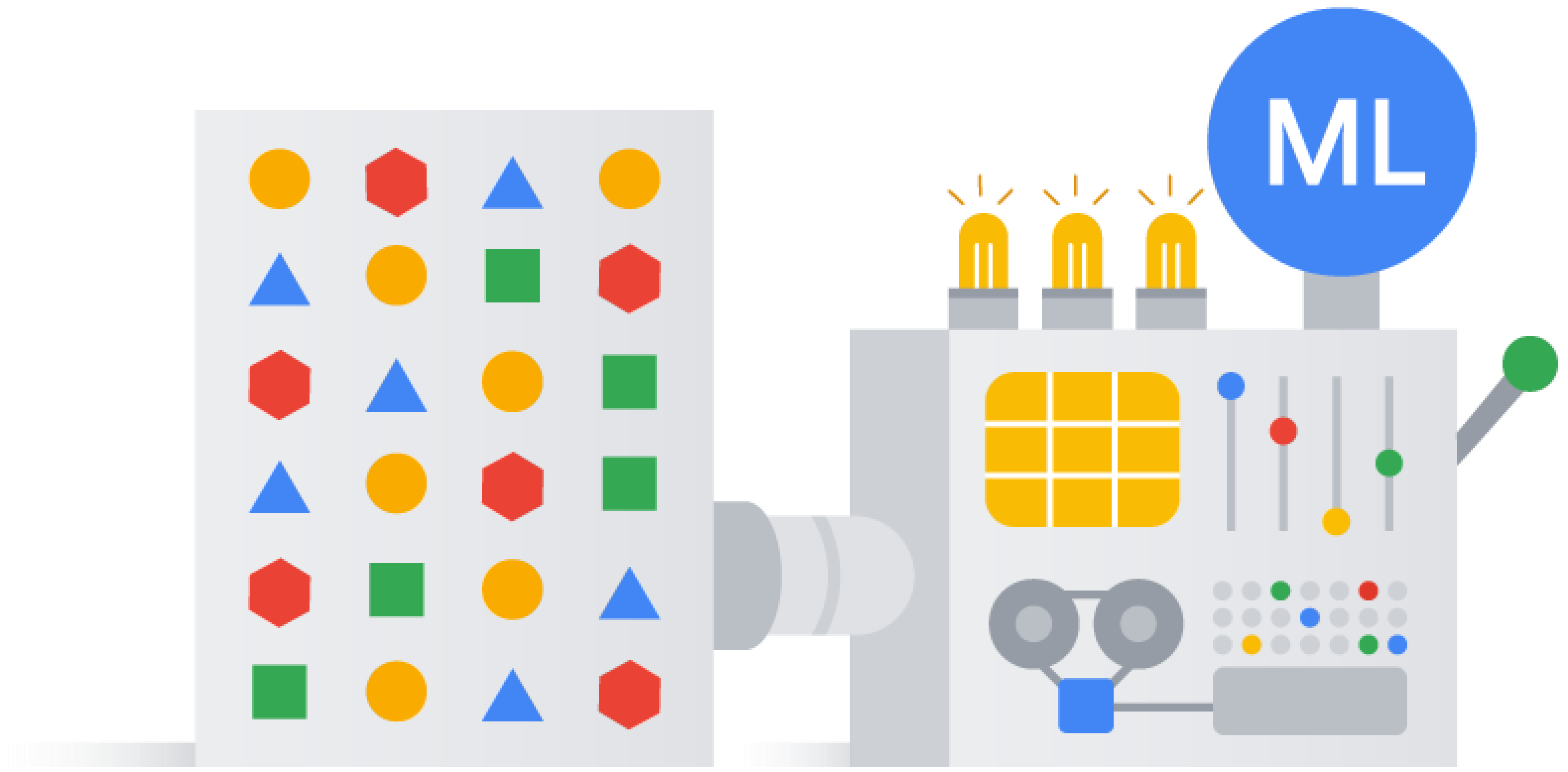
Parts in fractured condition

Predict with a trained model



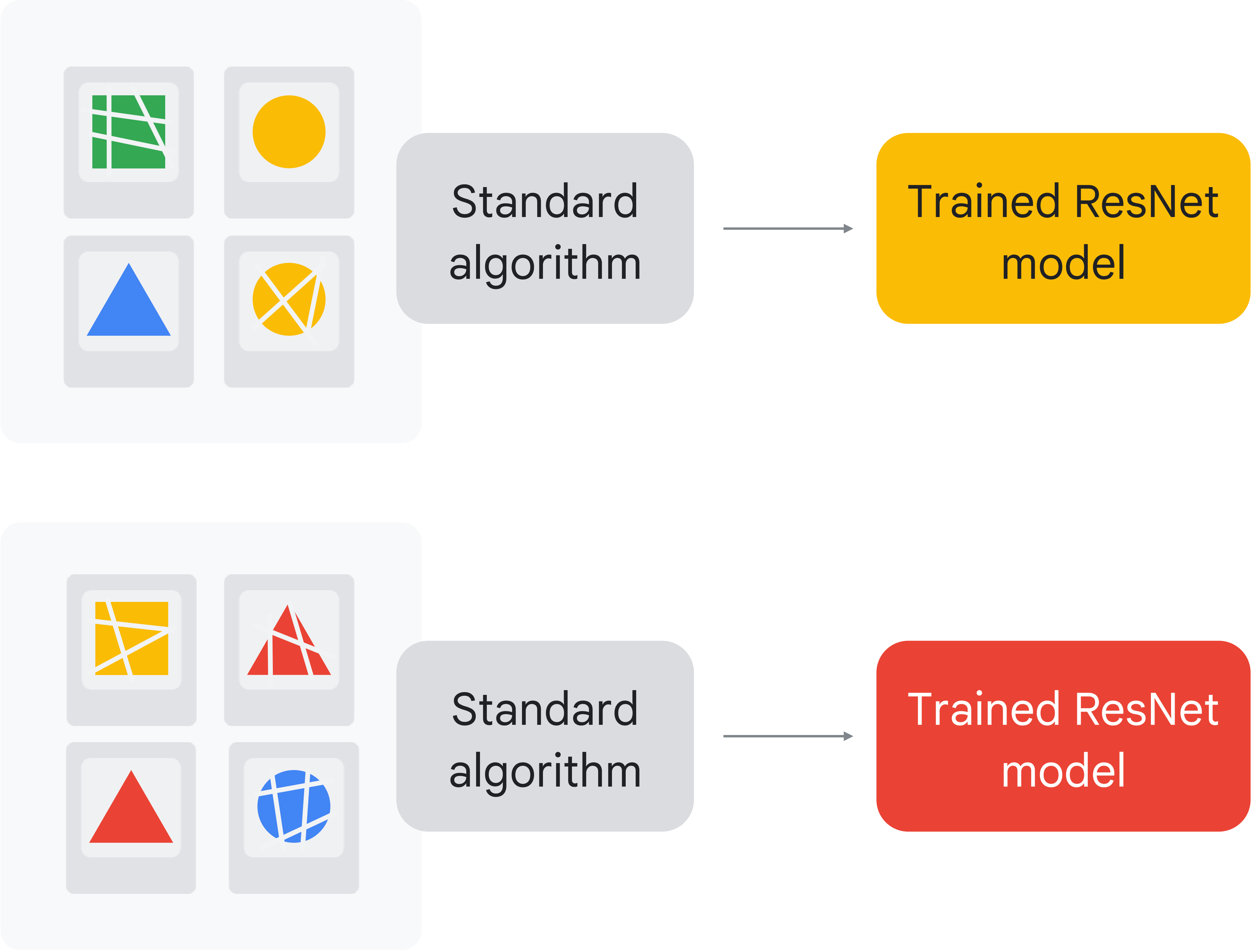
Standard algorithm use cases

- Detect a pattern in an image
- Predict the future of a time series
- Understand/transcribe human speech/text

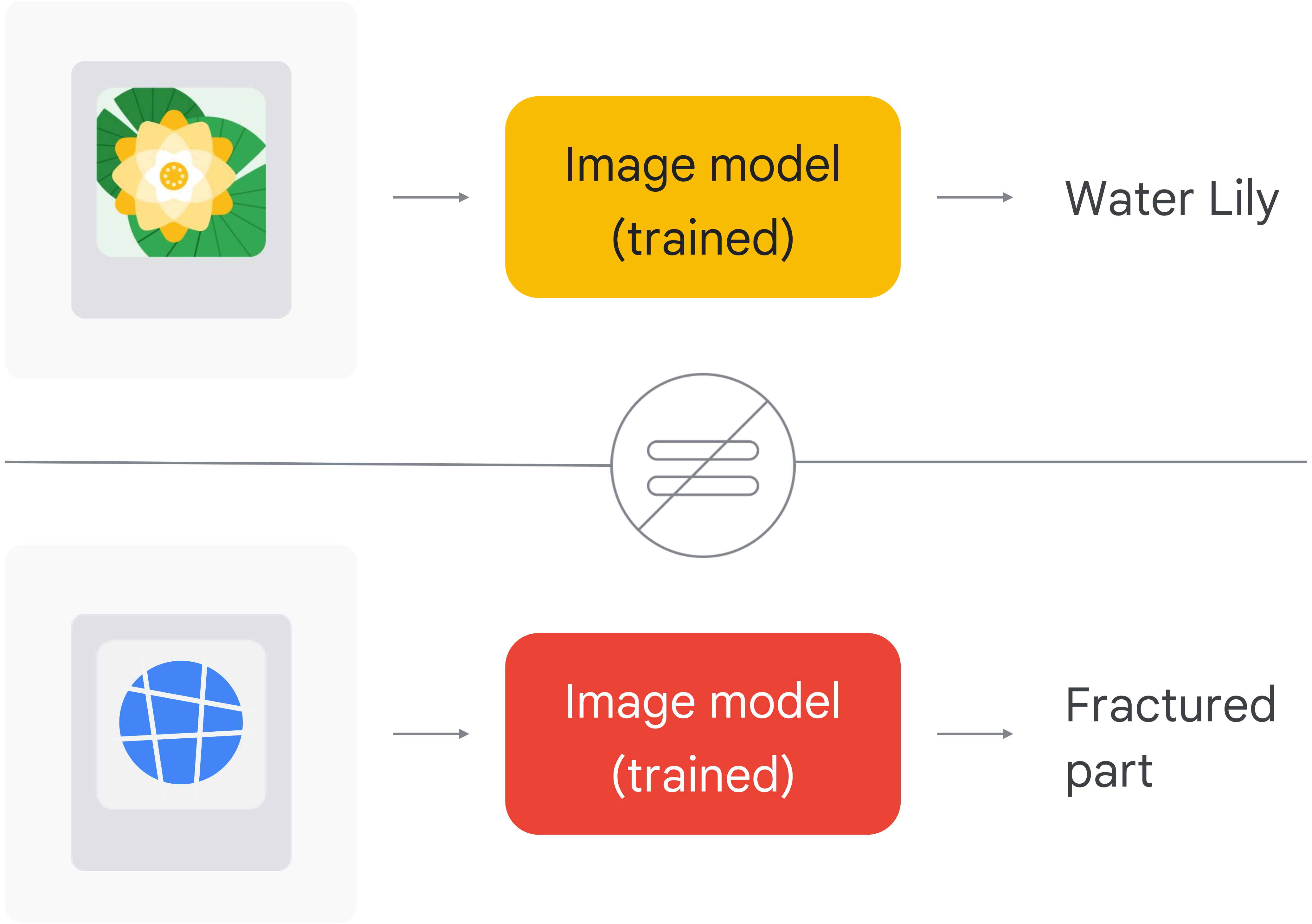


ResNet is a standard algorithm for image classification

The same algorithm applied to other data yields a different model



The algorithm is the same, but the trained model is different



Barriers to entry with ML have now fallen

Data
availability

Algorithm maturity
sophistication

Hardware
Software power

ML options on Google Cloud

01 BigQuery ML

Use SQL queries to create and execute machine learning models in BigQuery.

02 AutoML

A no-code solution to build ML models on Vertex AI.

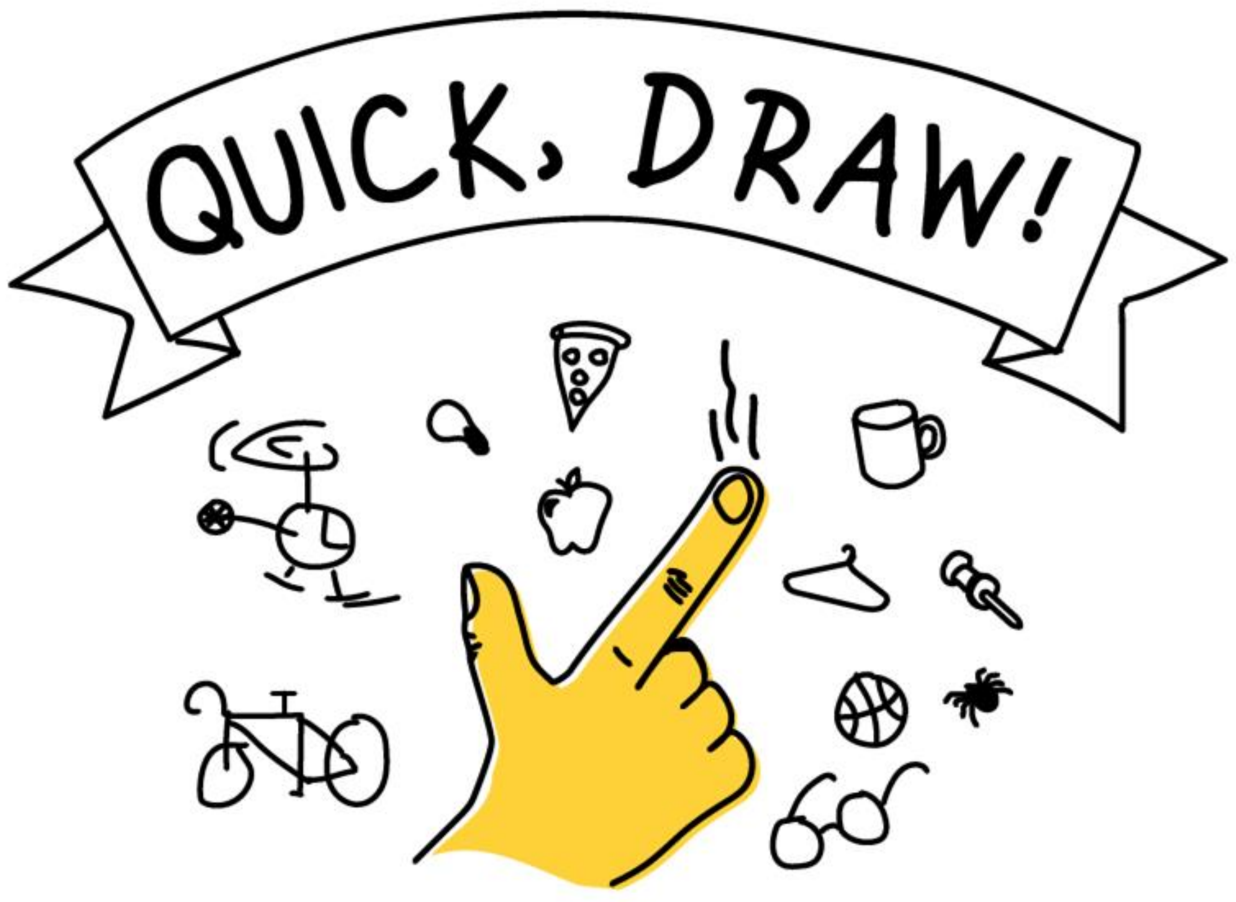
03 Custom training

Code your own ML environment to have the control over the ML pipeline.

04 Pre-built APIs

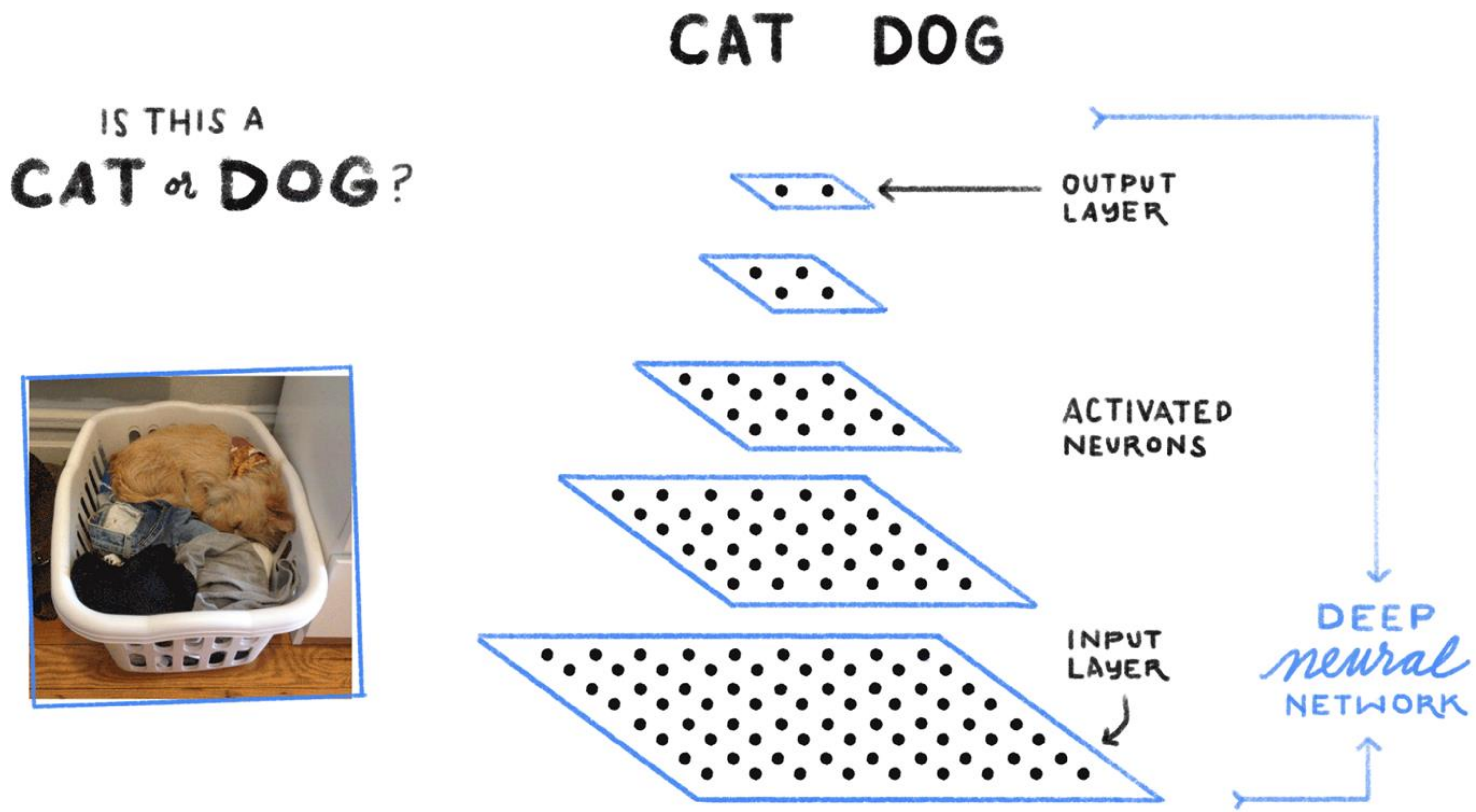
Leverage ML models that have already been built and trained by Google.

Having fun with ML: Quick, Draw

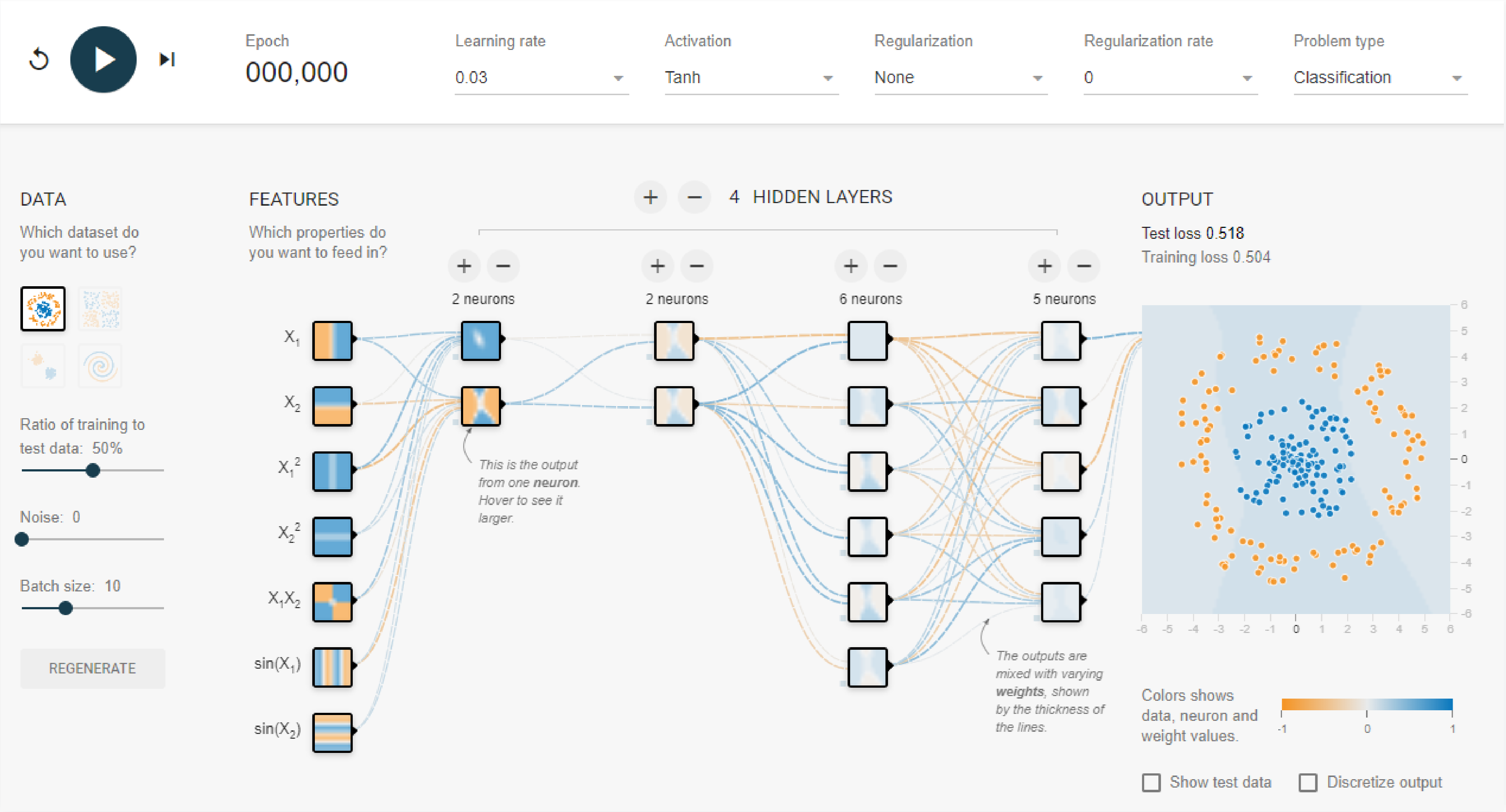


quickdraw.withgoogle.com

Modern AI applications use deep learning



The TensorFlow neural network playground



playground.tensorflow.org

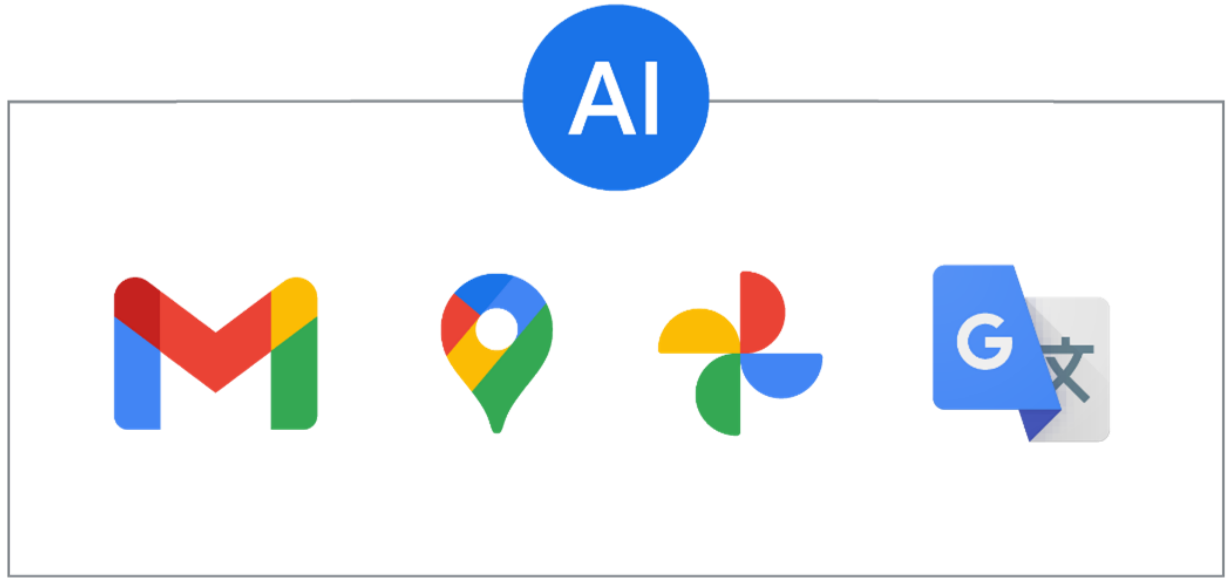
Agenda

Google Cloud

Google's AI technologies and products

2007	2015	2018	2019	2021
Scikit Learn	TensorFlow 1.0		TensorFlow 2.0	
Technology	Technology		Technology	
		AutoML BigQuery ML	AI Platform	Vertex AI
		Product	Product	Product

AI challenges: ML development



- Data
- Machine learning models
- Computing power

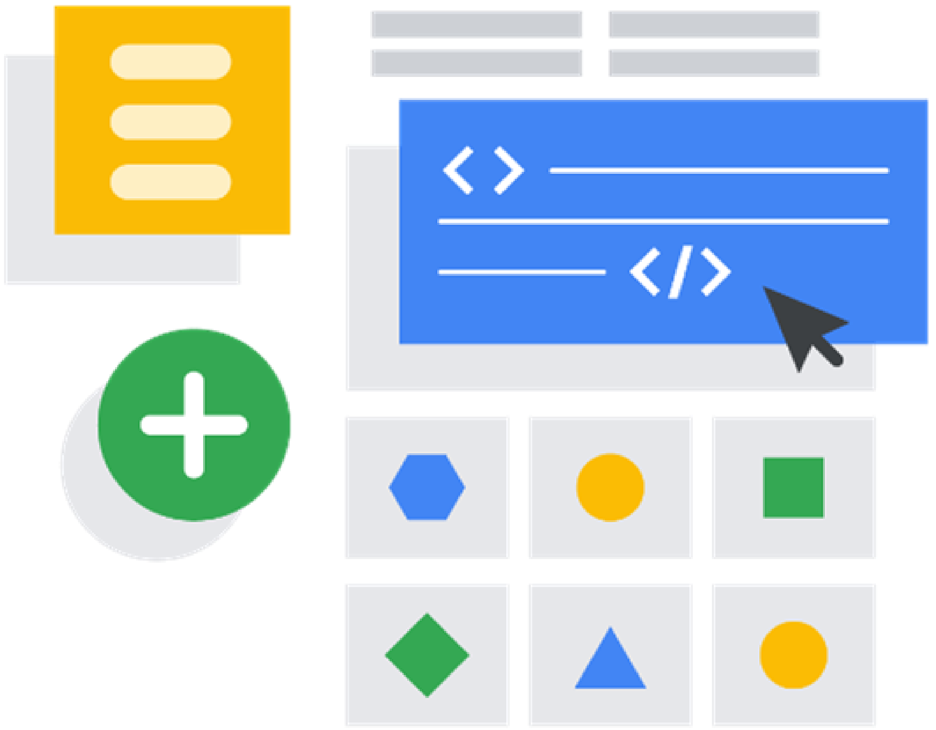
AI challenges: ML production

Only half of enterprise ML projects get past the pilot phase.

"Gartner Identifies the Top Strategic Technology trends for 2021"
- Gartner press release, October 19, 2020

- Scalability
- Monitoring
- Continuous integration and continuous delivery or deployment

AI challenges: ML tools ease-of-use



- Tools require advanced coding skills
- Take focus away from model configuration
- No unified workflow
- Difficulties finding tools

Google's solution is Vertex AI

Production challenges

Vertex AI

Ease-of-use challenges

Vertex AI is a unified platform

Create models | Manage models | Deploy models

Data readiness

Users can upload data from wherever it's stored–Cloud Storage, BigQuery, or a local machine.

Feature readiness

Users can create features, which are the processed data that will be put into the model, and then share them with others by using the feature store.

Training and hyperparameter tuning

When the data is ready, users can experiment with different models and adjust hyperparameters.

Deployment and model monitoring

Users can set up the pipeline to transform the model into production by automatically monitoring and performing continuous improvements.

Vertex AI provides two solutions in one

Vertex AI

- AutoML: no-code solution
- Custom training: code-based solution

The benefits of Vertex AI

Seamless

Scalable

Vertex AI

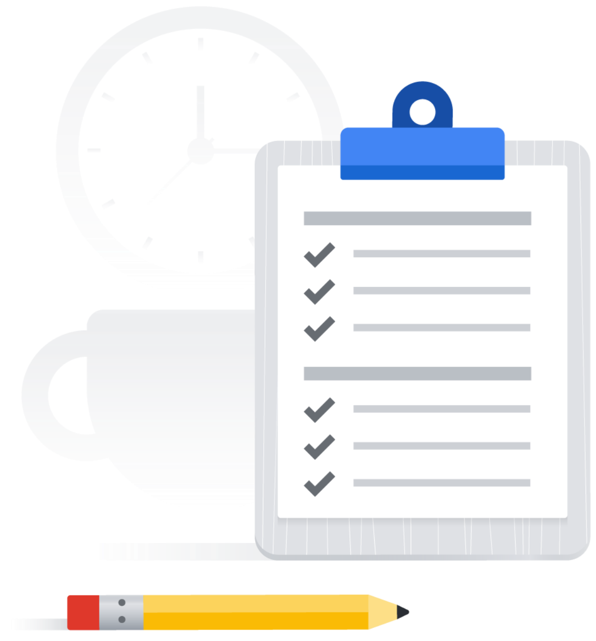
Sustainable

Speedy

Agenda

Lab: Vertex AI: Qwik Start

Hands-on lab



During this lab you'll:

1. Train a TensorFlow model locally in a hosted Vertex Notebook.
2. Create a managed Tabular dataset artifact for experiment tracking.
3. Containerize your training code with Cloud Build and push it to Google Cloud Artifact Registry.
4. Run a Vertex AI custom training job with your custom model container.
5. Use Vertex TensorBoard to visualize model performance.
6. Deploy your trained model to a Vertex Online Prediction Endpoint for serving predictions.
7. Request an online prediction and explanation and see the response.

Agenda

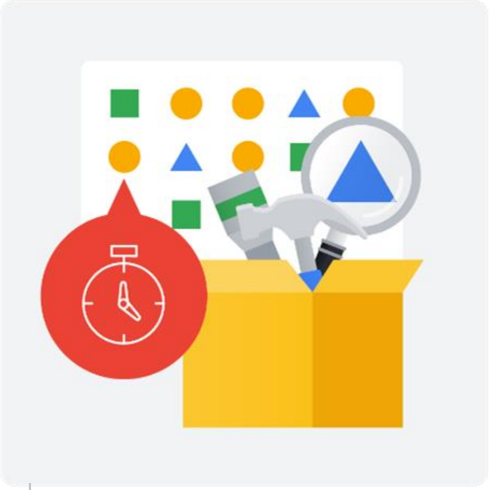
Google Cloud

AutoML

01	02	03	04
BigQuery ML	AutoML	Custom training	Pre-built APIs
Use SQL queries to create and execute machine learning models in BigQuery.	A no-code solution to build ML models on Vertex AI.	Code your own ML environment to have the control over the ML pipeline.	Leverage ML models that have already been built and trained by Google.

Traditional ML development vs. AutoML

Traditional ML development

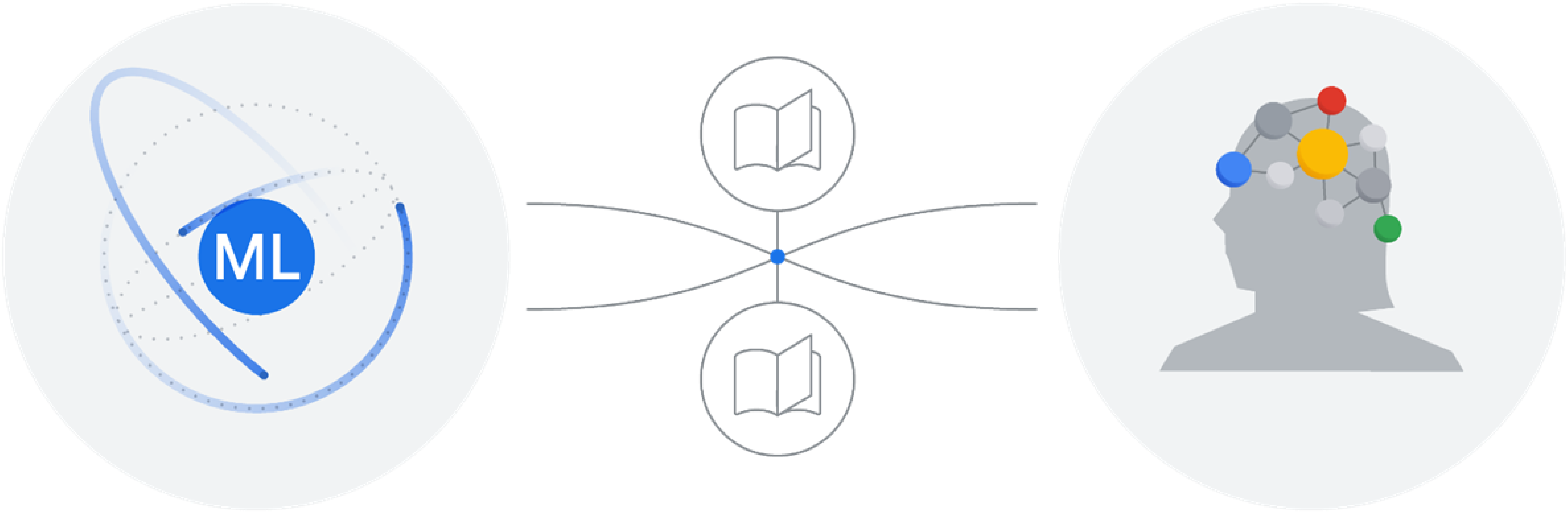


- Repeatedly add new data and features
- Try different models
- Tune hyperparameters

AutoML: Automated machine learning (2018)

- Automates machine learning pipelines to save data scientists from manual work

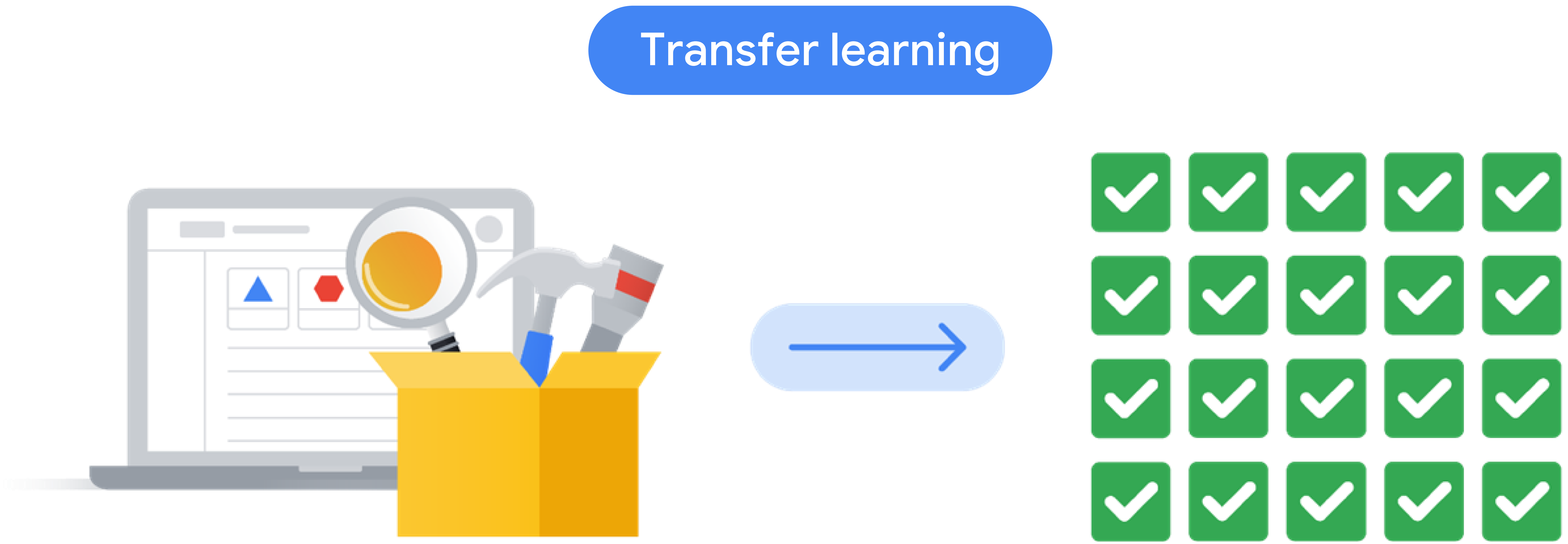
Machine learning is similar to human learning



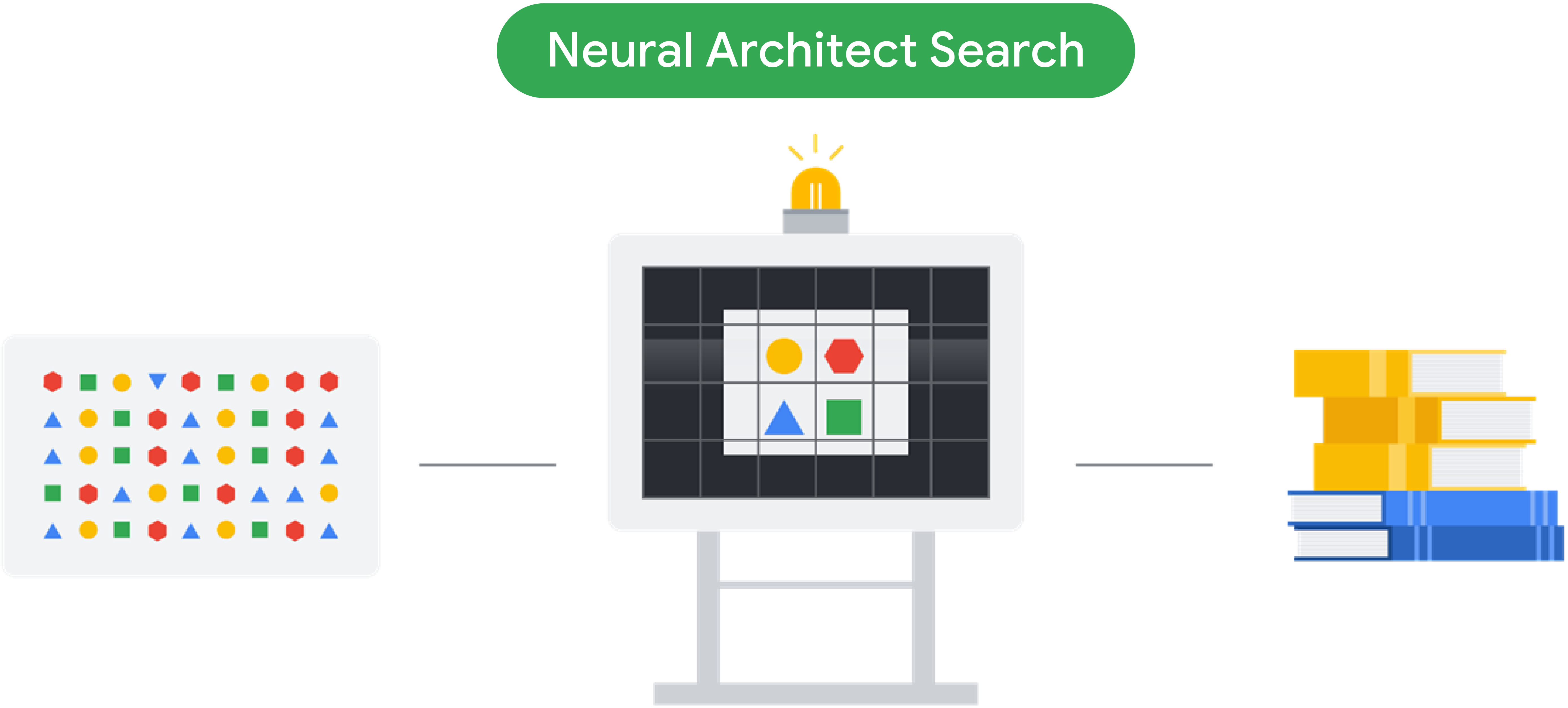
Transfer learning

Build a knowledge base in the field

Transfer learning from pre-trained models



Neural Architect Search finds the optimal model



AutoML

- Powered by the latest ML research
- Trains and evaluates multiple models
- Produces an ensemble of ML models
- Chooses the best one

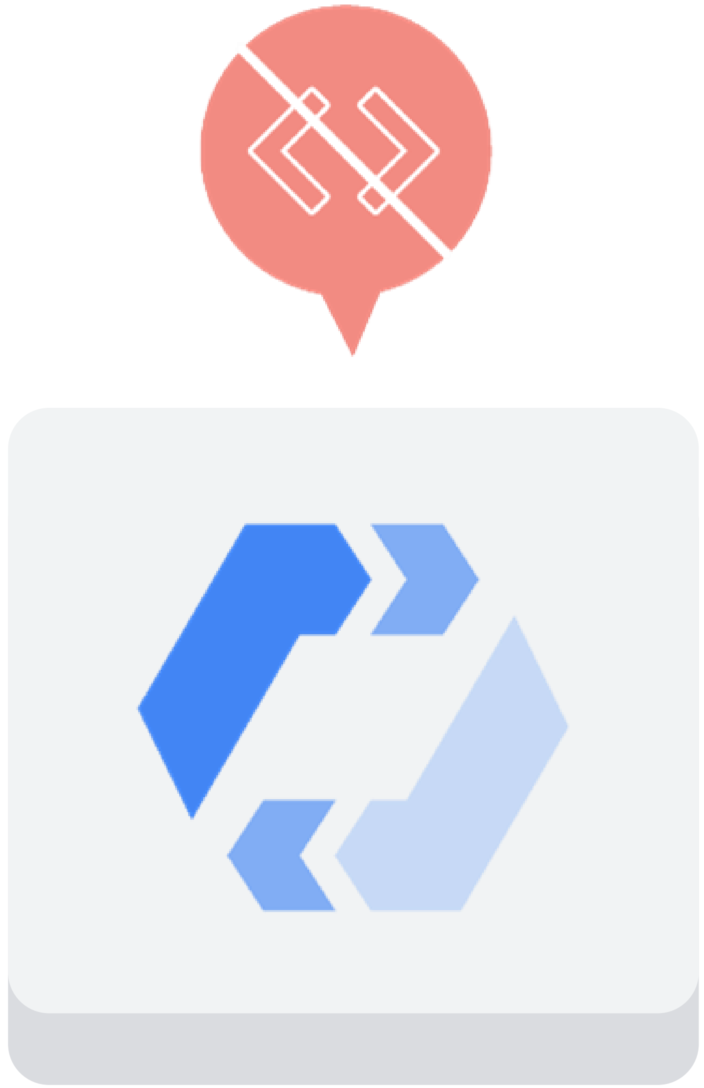
Benefits of AutoML

Train custom ML models with

- Minimal effort
- Little machine learning expertise

Allows data scientists to focus on

- Defining business problems
- Evaluating and improving model results



AutoML supports four types of data

Image

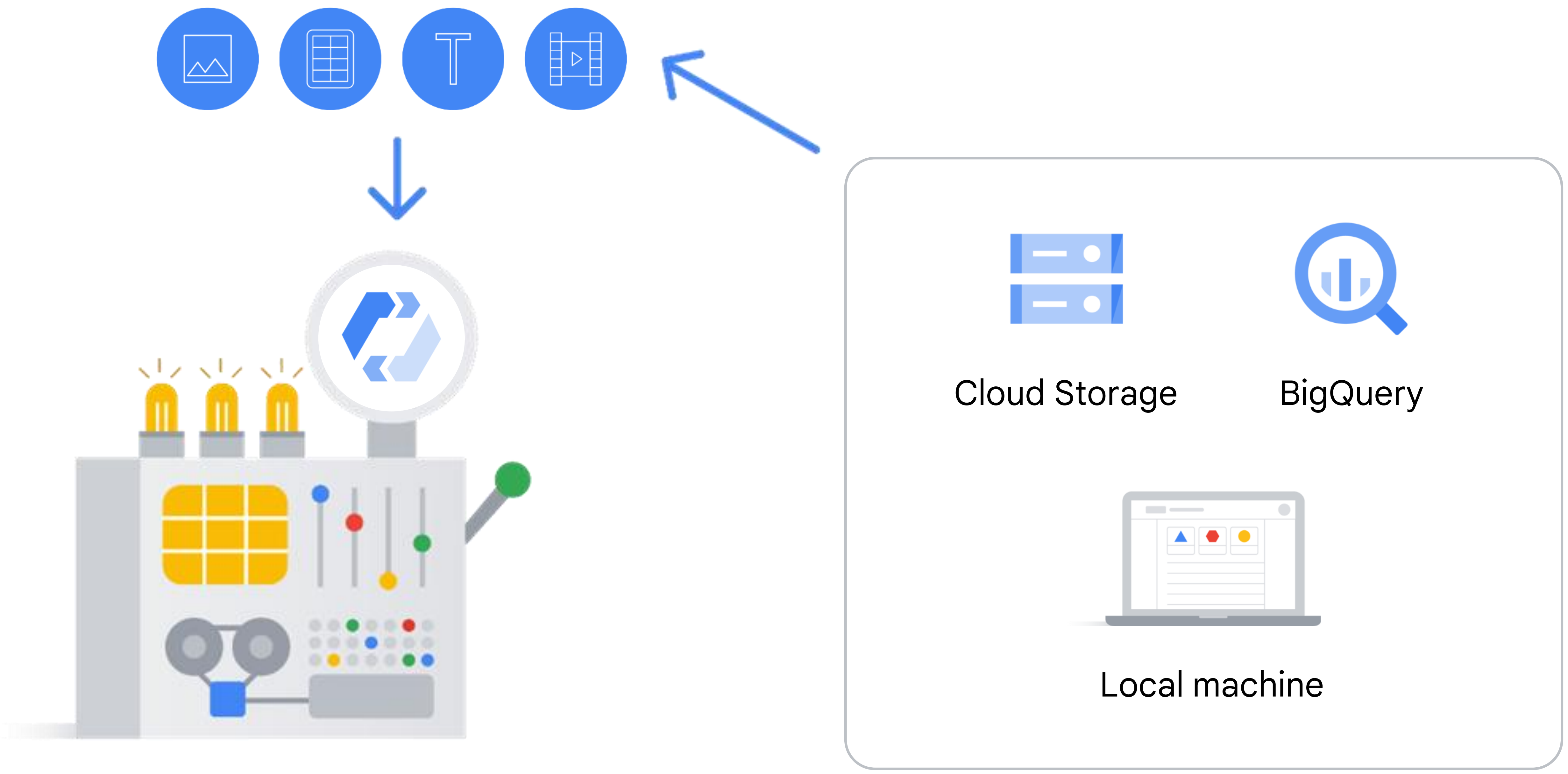
Tabular

Text

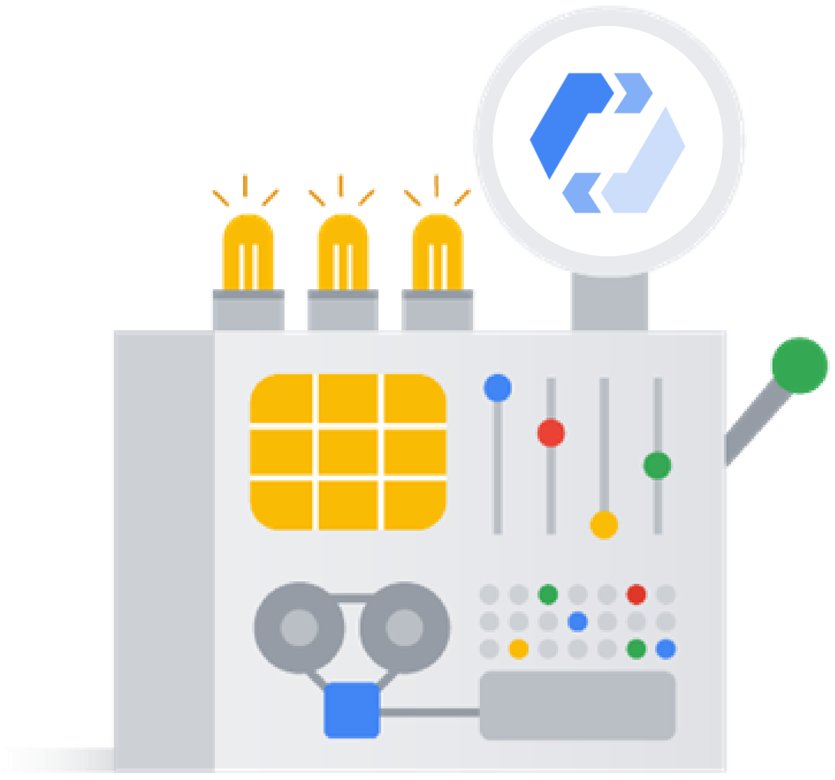
Video

AutoML solves different types of problems, called **objectives**

To start, upload your data into AutoML



Inform AutoML of the problems you want to solve



Inform AutoML of problems to solve

May sound similar to pre-built APIs

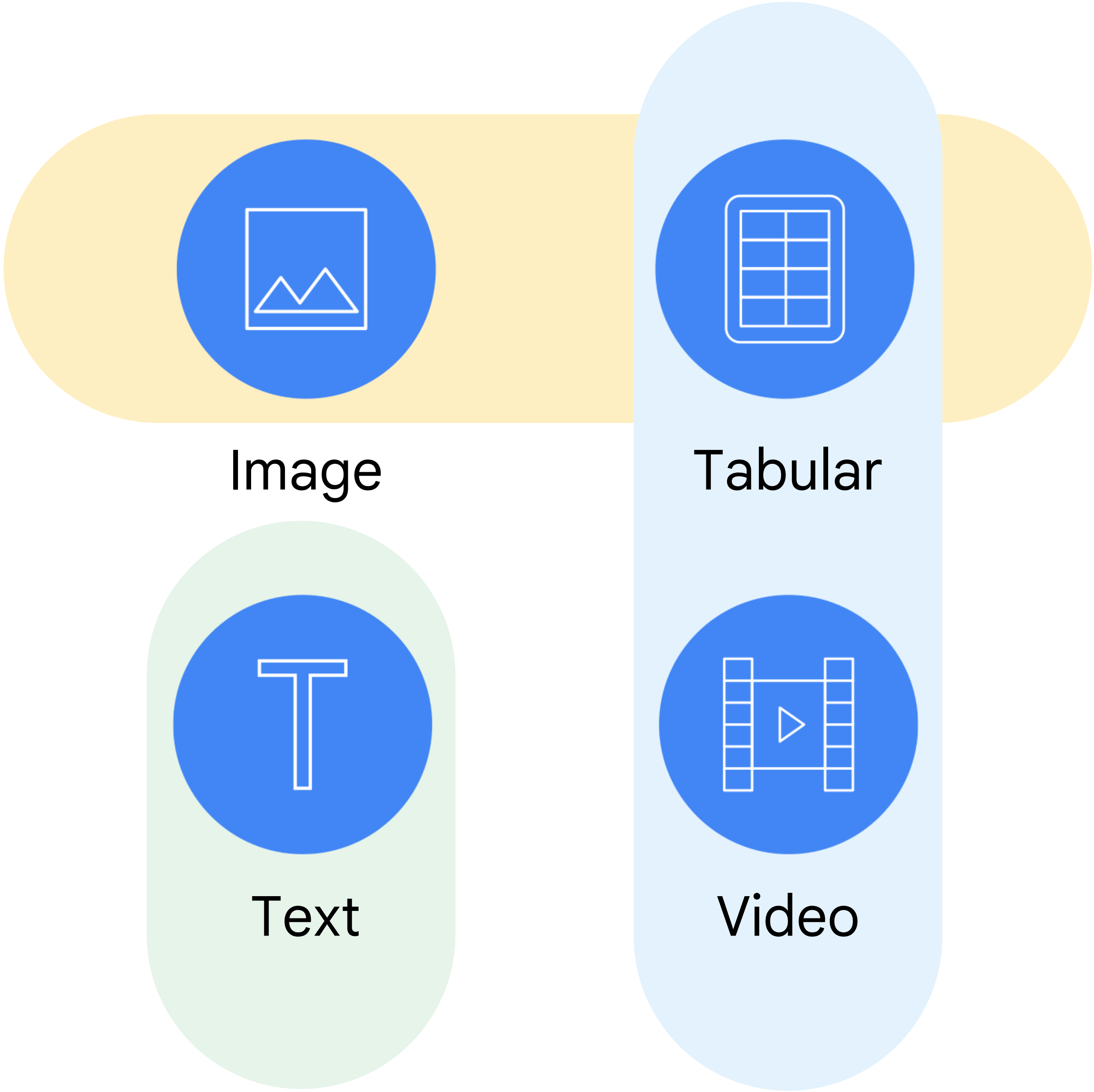
Pre-built APIs

Use pre-built ML models

AutoML

Uses custom-built ML models

AutoML solves mixed data types and objectives



Agenda

AutoML

01

BigQuery ML

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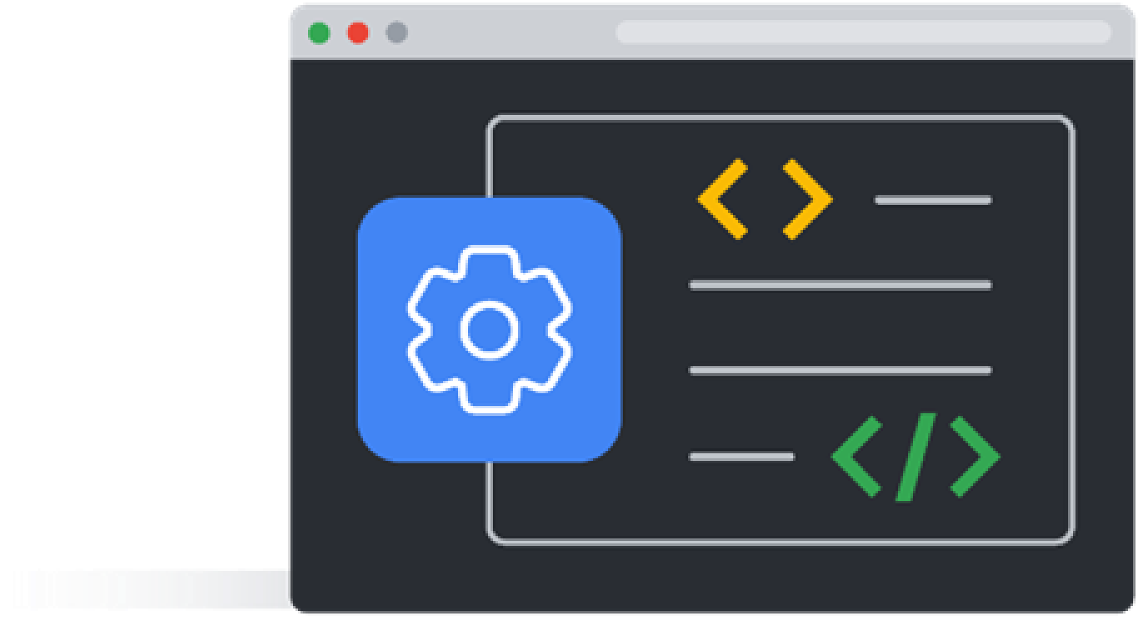
Leverage ML models that have already been built and trained by Google.

Vertex AI Workbench

Single development environment

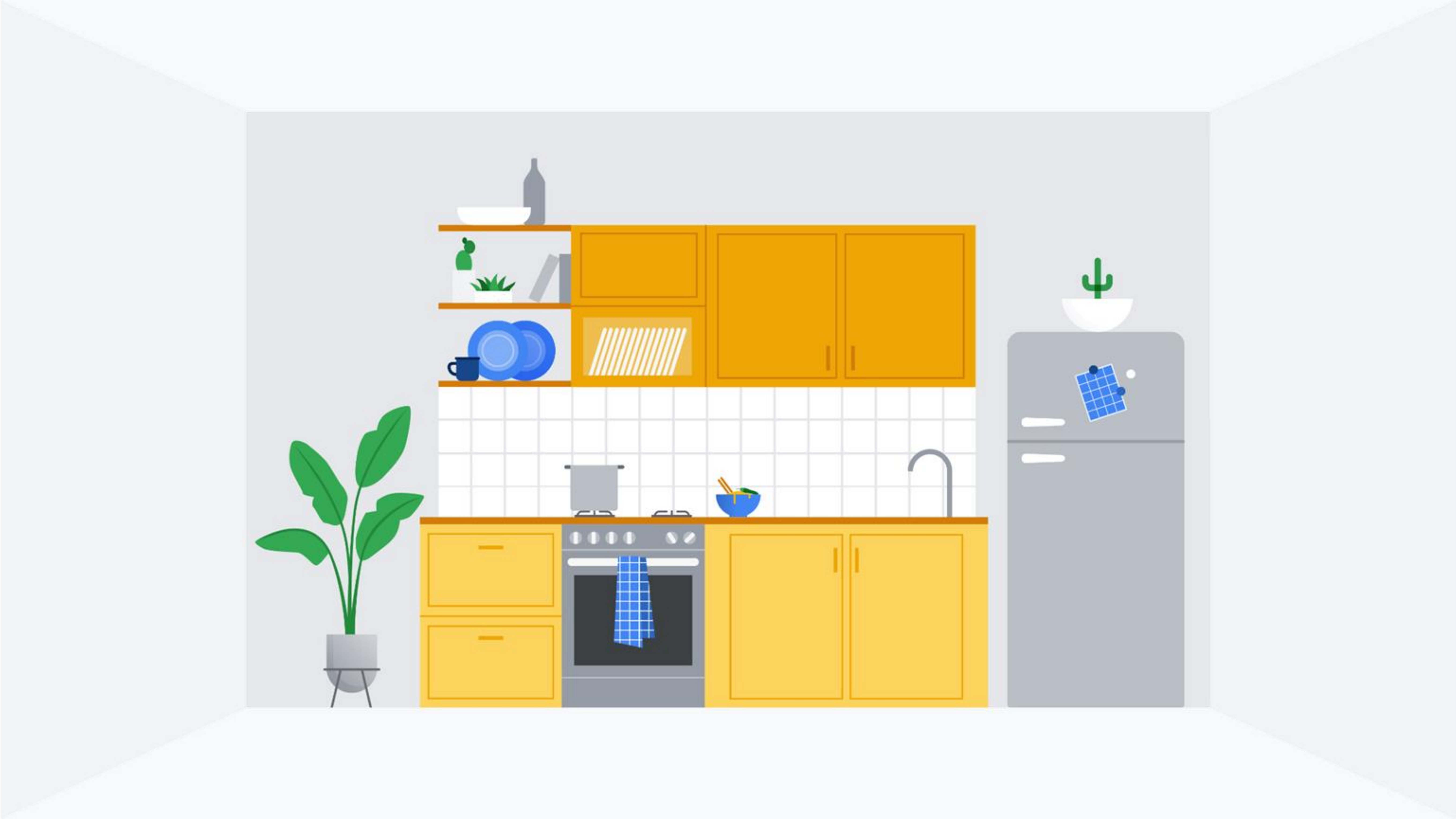
Vertex AI Workbench

ML training code environment options

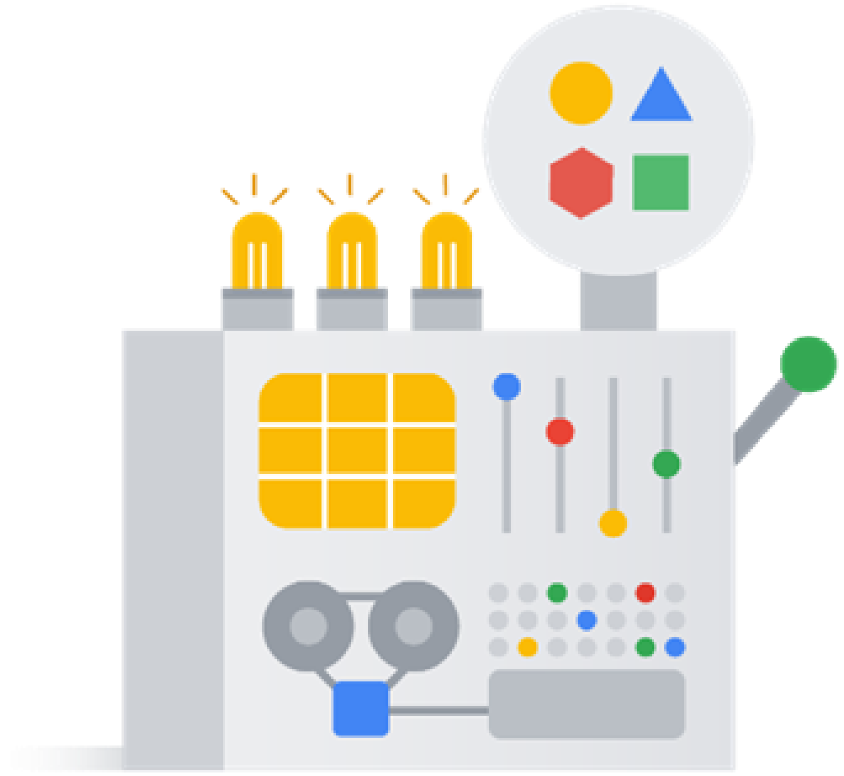


- A pre-built container
- A custom container

A pre-built container is like a fully furnished room



Pre-built containers: Pre-installed environment



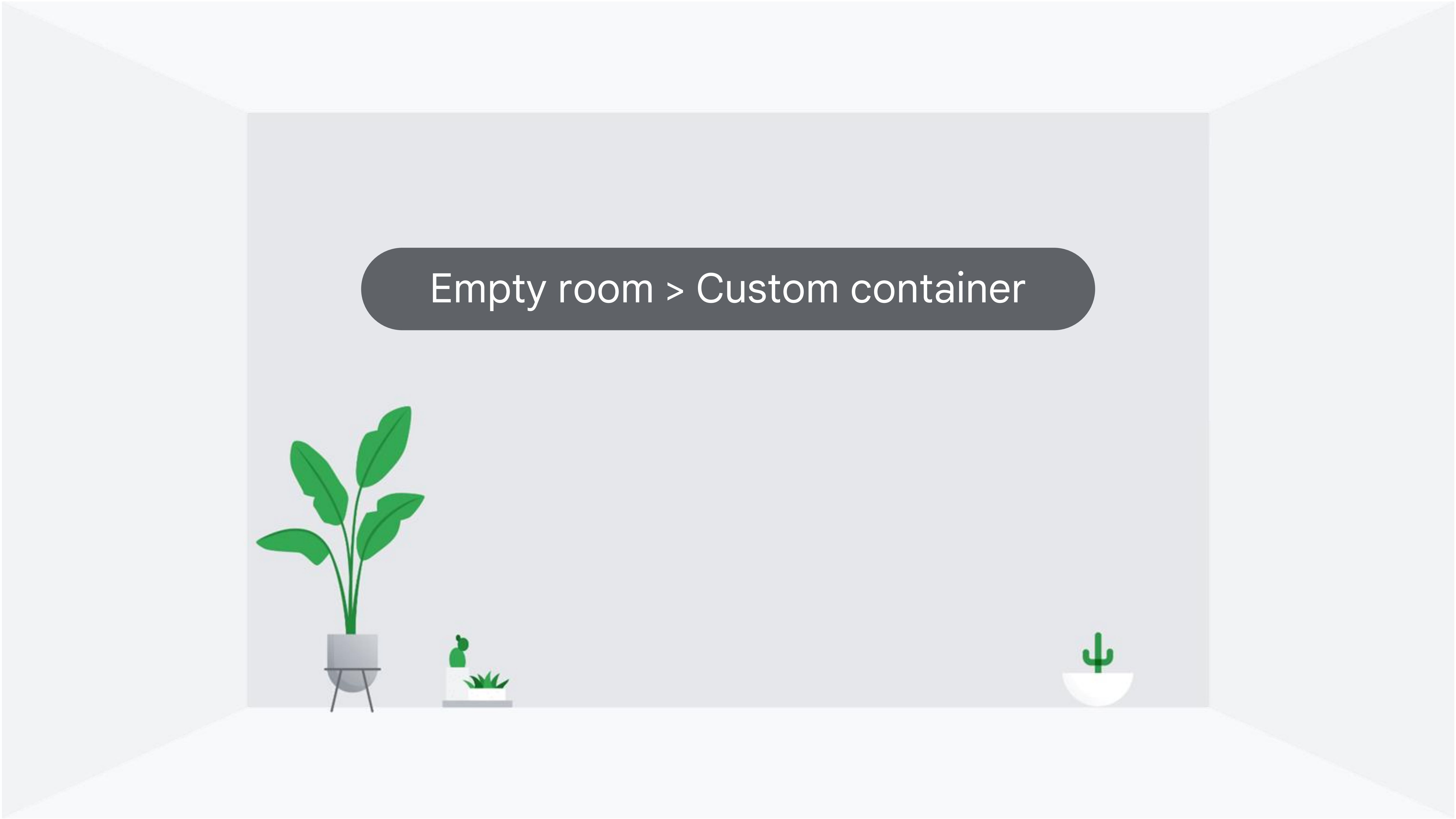
→

- TensorFlow
- Pytorch
- Scikit-learn
- XGboost
- Python code

→

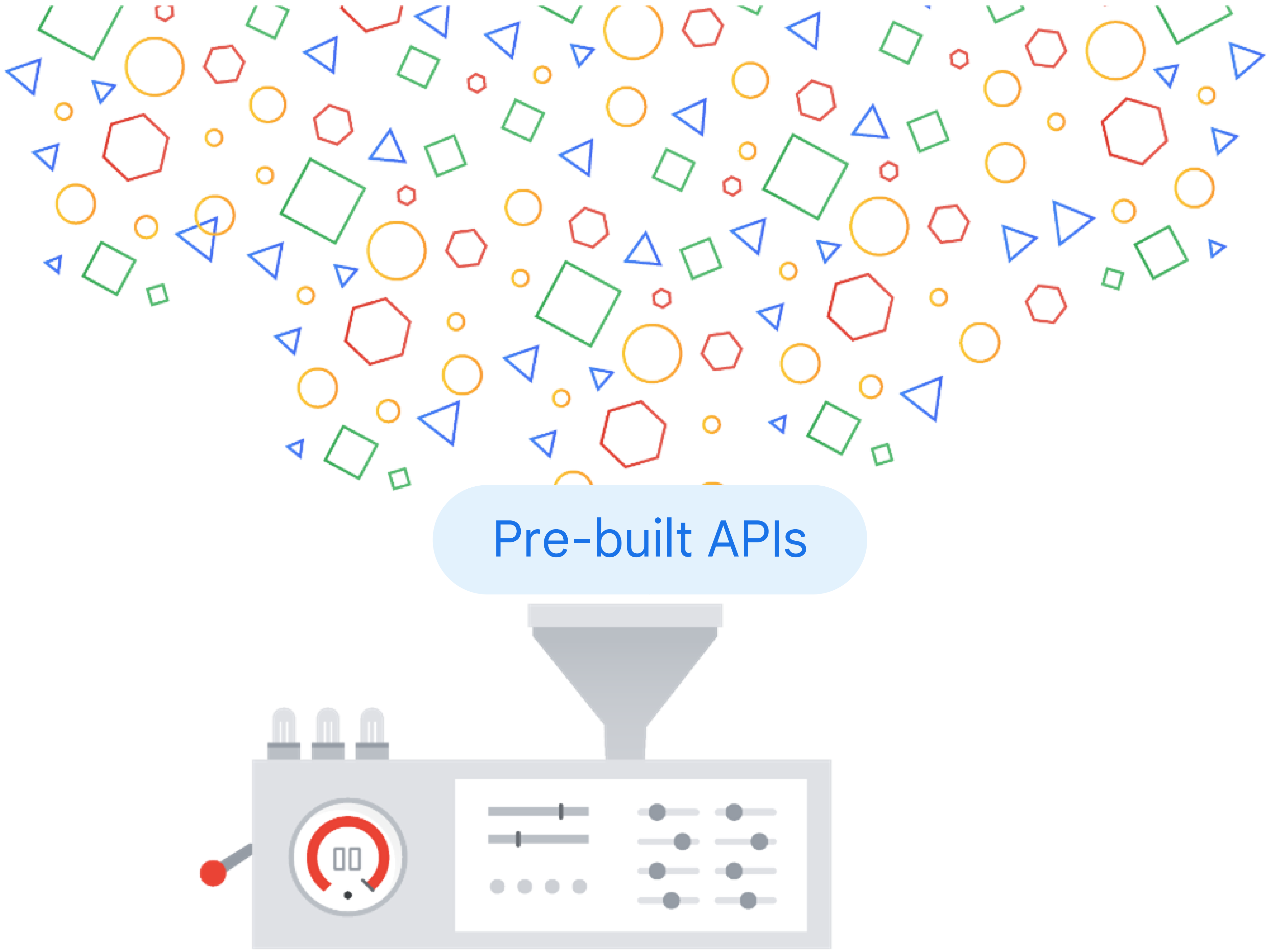
Pre-built container

A custom container is like an empty room



Agenda

Pre-built APIs don't require custom training data



Offered as services

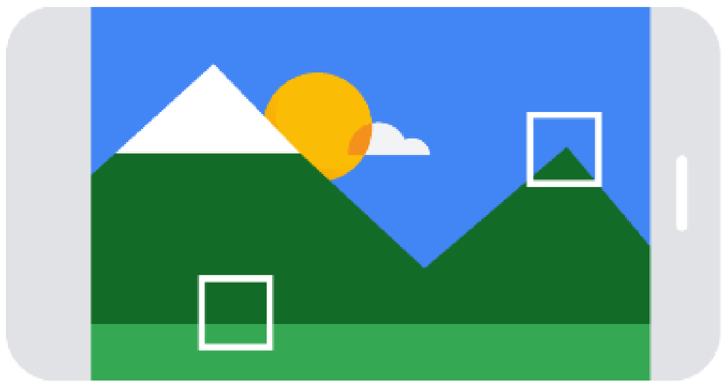
- Build datasets
- Curate datasets
- Traine datasets

Pre-built API examples

Speech-to-Text API	Converts audio to text for data processing.
Cloud Natural Language API	Recognizes parts of speech called entities and sentiment.
Cloud Translation API	Converts text from one language to another.
Text-to-Speech API	Converts text into high-quality voice audio.
Vision API	Works with and recognizes content in static images.
Video Intelligence API	Recognizes motion and action in video.

Pre-built APIs trained with Google datasets

Vision API



Based on

Google’s image datasets

Speech-to-Text API

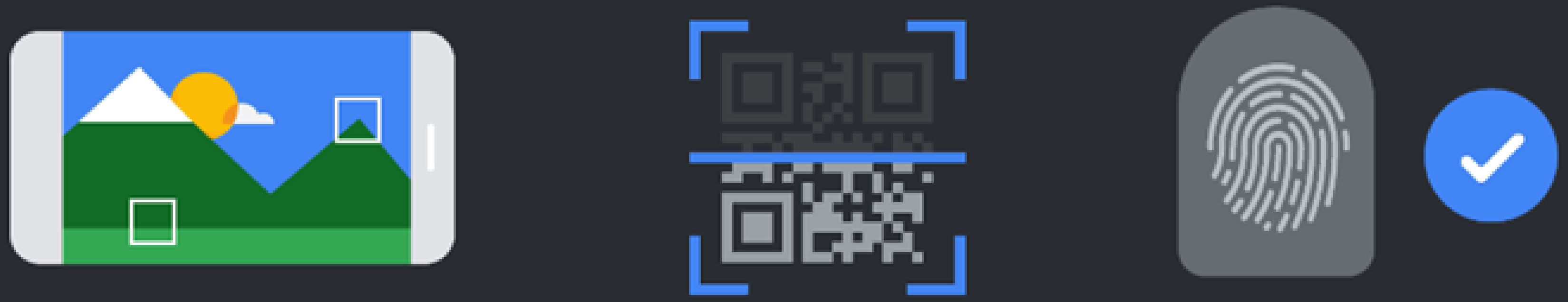
Trained on

YouTube captions

Translation API

Built on

Google Translate



Vision API

cloud.google.com/vision

DEMO

Try the API

Try the API

Labels | Properties | Safe Search

PXL_20210928_233109458.jpg

Wood	88%
Creative Arts	82%
Art	81%
Flooring	81%
Hardwood	75%
Wood Stain	65%
Plywood	65%
Visual Arts	62%

Show JSON

RESET

NEW FILE


Google Cloud + Bloomberg

Agenda

Cloud Natural Language API: Qwik Start

Hands-on lab

During this lab you'll:

1. Create an API key.
2. Make an entity analysis request.

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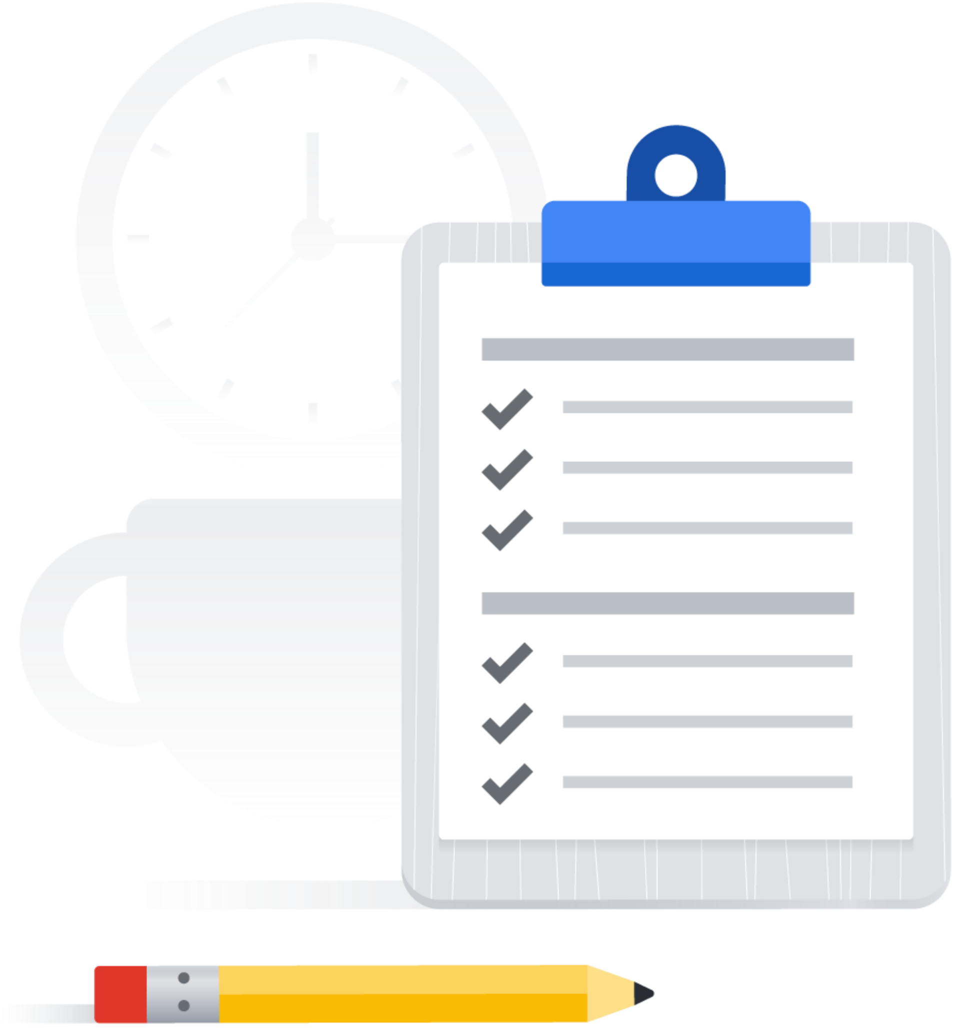
Google Cloud Speech API: Qwik Start

Hands-on lab

During this lab you'll:

1. Create an API key.
2. Create a Speech API request.
3. Call the Speech API request.



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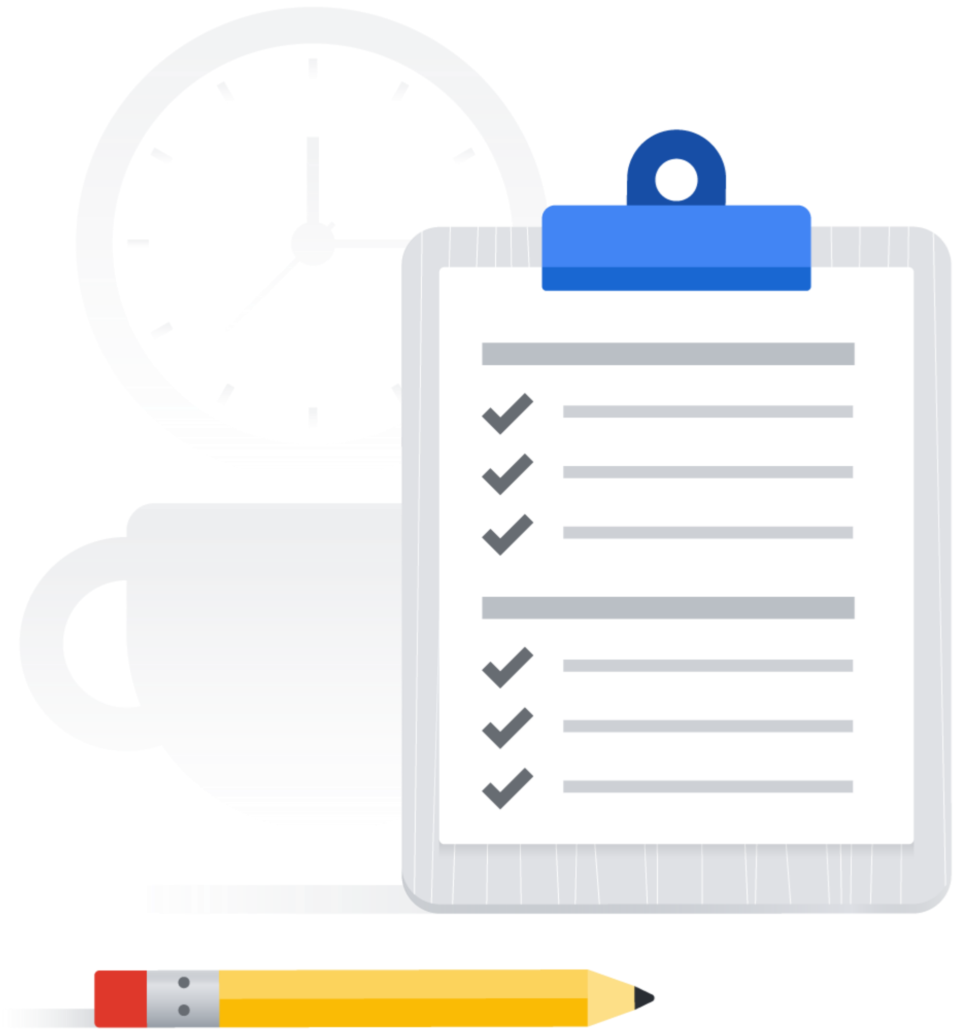
Video Intelligence: Qwik Start

Hands-on lab

During this lab you'll:

1. Enable the Video Intelligence API.
2. Set up authorization.
3. Make an annotated video request.



Agenda

Knowledge check

Which of these is a no-code solution that lets you build your own machine learning models on Vertex AI through a point-and-click interface?

A. BigQuery ML

B. Pre-built APIs

C. AutoML

D. Custom training

Knowledge check

Which of these is a no-code solution that lets you build your own machine learning models on Vertex AI through a point-and-click interface?

A. BigQuery ML

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C. AutoML

D. Custom training

Knowledge check

What is the name of Google's unified AI platform that brings all the components of the machine learning ecosystem and workflow together?

A. TensorFlow

B. Vertex AI

C. Cloud AI

D. AutoML

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What is the name of Google's unified AI platform that brings all the components of the machine learning ecosystem and workflow together?

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Knowledge check

What Google machine learning API can be used to convert audio to text for data processing?

A. Cloud Natural Language API

B. Speech-to-Text API

C. Vision API

D. Video Intelligence API

Knowledge check

What Google machine learning API can be used to convert audio to text for data processing?

A. Cloud Natural Language API

B. Speech-to-Text API

C. Vision API

D. Video Intelligence API

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Summary

- Defined machine learning (ML), explored the terminology used, and identified the value proposition.
- Explored Vertex AI, Google's unified AI platform.
- Learned about Google APIs to apply a range of pre-trained ML models.